Carbon Cycle Interactive

https://www.windows2universe.org/earth/climate/carbon_cycle.html

1. How does carbon exist in the atmosphere?
   _____
2. Which two possible places can carbon go from the atmosphere?
   _____
3. Which two possible places can carbon go from land plants?
   _____
4. Which three possible places can carbon go from the surface ocean?
   _____
5. Which two possible places can carbon go from marine life?
   _____
6. *What living things are missing from this diagram? _____

Land plants: You have been taken out of the atmosphere by a plant through a process called _____. As more carbon dioxide is added to our atmosphere, plants will be able to grow _____. Plants also release carbon back to the atmosphere by _____.

Soil: You are now a part of the soil called _____, which is decomposing plants and animals. Soil is also made of inorganic parts such as sand, silt, and clay. Soils store about 3% of Earth's carbon. As bacteria and fungi breakdown the detritus, carbon is sent into the _____.

*what would this matter turn into if it was left and compressed into the soil?

Surface ocean: Either you got here by _____ from the atmosphere, by _____ marine life, or from circulating water from the deep ocean. The ocean absorbs _____ carbon dioxide from the atmosphere than the land does. The surface ocean takes in approximately _____ of carbon per year. Cold water absorbs carbon _____ than warm water.

Marine Life: Tiny marine organisms called _____ take in carbon to make the nutrition they need through a process called _____. The phytoplankton are eaten by larger marine life. Marine life cannot survive without carbon, but high levels of carbon dissolved in ocean waters are harmful to marine organisms such as algae, mollusks and corals.

Deep Ocean: The deep ocean gets carbon from circulation with the surface ocean and dead and decaying marine life. When carbon gets to the deep ocean, it usually stays there for _____ before moving on. The deep ocean holds more than _____ of the Earth's carbon.

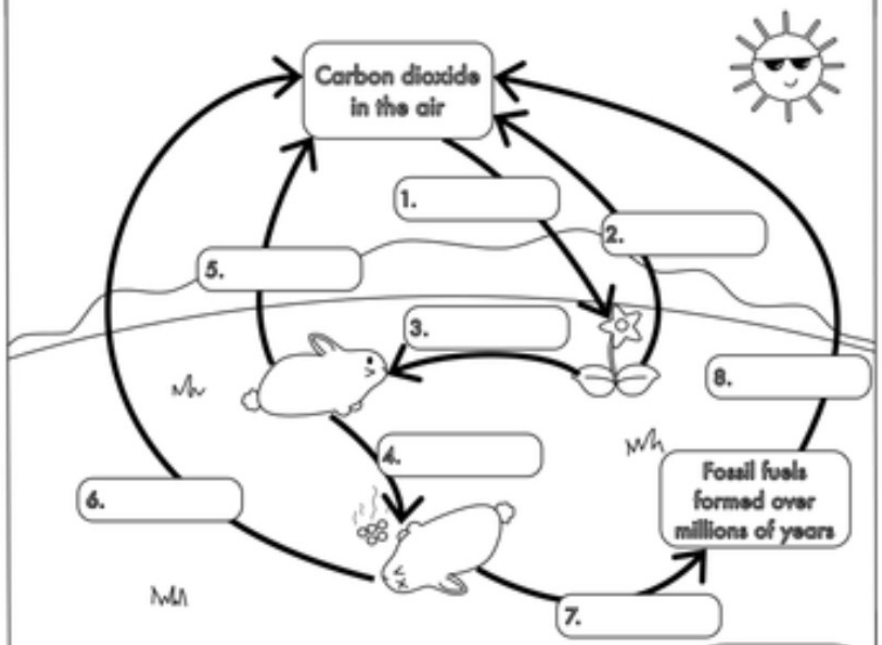
Then play the nitrogen cycle game: https://www.brainpop.com/games/nitrogencyclegame/

The Carbon Cycle

Label the numbered boxes correctly with the following:

Respiration by animals | Feeding | Decomposition | Death and waste

No decomposition | Combustion | Respiration by plants | Photosynthesis



Key Points

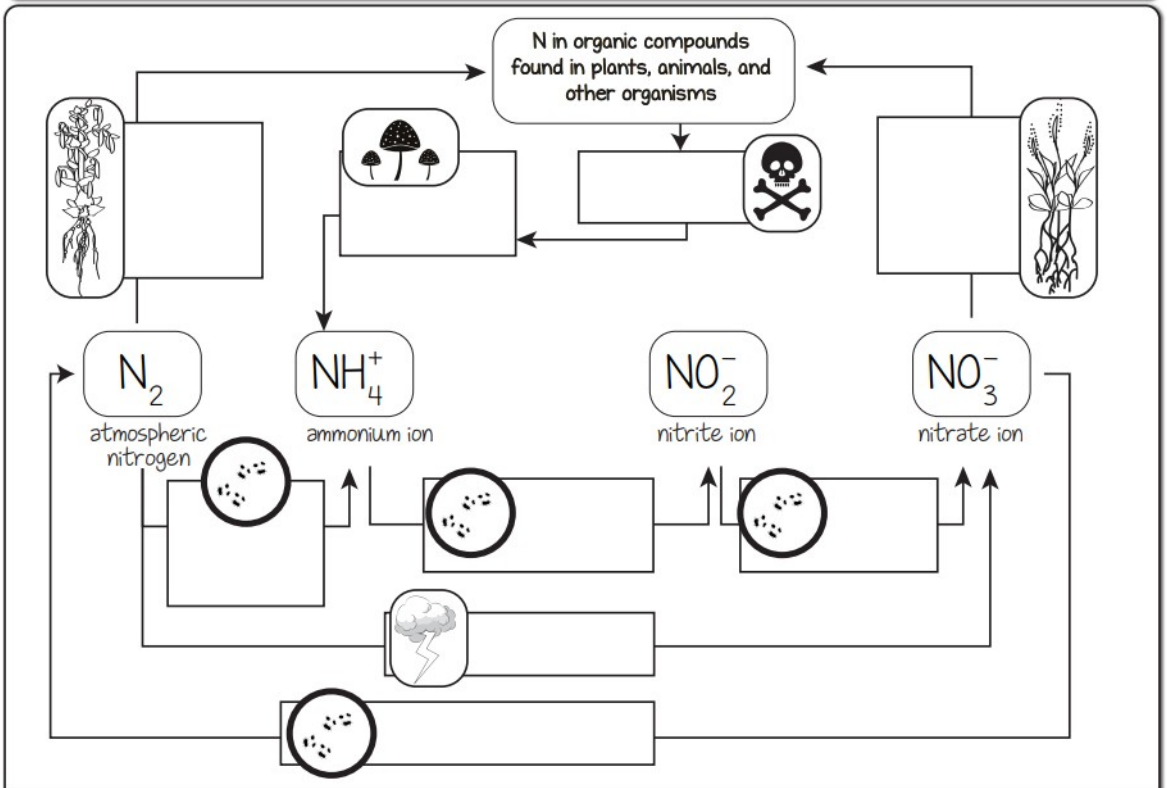
Words to use
respiration dioxide
decomposers carbon
organisms fungi

- The carbon cycle involves the cycling of carbon between the environment and _ _ _ _ _ _ _ _ _ _.
- Photosynthesis and _ _ _ _ _ _ _ _ _ _ _ _ are the key processes involved in the cycle.
- Also decomposition which involves microorganisms called _ _ _ _ _ _ _ _ _ _ _ _ is important to digest waste and dead organisms. These microorganisms are bacteria and _ _ _ _ _.
- These decomposers carry out respiration which coverts _ _ _ _ _ _ compounds in dead and waste material back to carbon _ _ _ _ _ _ _ in the air.

Name: Class: Date:

Nitrogen Cycle

Using the wordbank below, fill in the boxes to describe the players in the nitrogen cycle



1. decomposition by fungi and bacteria
2. lightning
3. most plant uptake
4. dead organisms and waste
5. nitrogen fixing bacteria in legume root nodules
6. nitrogen fixing bacteria in soil
7. denitrifying bacteria
8. ammonium oxidizing bacteria
9. nitrite oxidizing bacteria